Legal Protection of Digital Property

Lecture 2

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Software Copyright in HK

- governed by the Copyright Ordinance (Cap 528)

NB. Discussions on copyright in this course concern only the economic rights of the copyright owner. The moral rights of the author (and film director) are not dealt with but can be found in CO, ss.89-100.

Protected Subject Matter

1. Source codes and object codes
   - CO, s.4(1): computer programs are protected as literary works
   - “computer program” not defined in the Ordinance but no dispute that it includes source codes, which are human-readable
   - one argument against object codes being afforded literary copyright: not human readable, therefore not "literary" works

- now clear that object codes are protected by virtue of TRIPS, Art 10(1):
  - » "Computer programs, whether in source or object code, shall be protected as literary works under the Berne Convention". As a member of WTO, HK must give effect to TRIPS: Art 1(1).

NB. Even without TRIPS, object code is protected in HK indirectly via source code as an "adaptation" of the latter (see lecture 3).

2. Algorithms

- steps for the solution of a specific problem independent of the code in which such steps will be expressed

- algorithm vs program:
  - » an algorithm is expressed in **natural language** and denotes the steps for solving a problem whereas a program is an implementation of such steps using a specific **programming language**
  - » related to idea/expression dichotomy – is algorithm idea? Or expression?

- *Ibcos Computers v Barclays Mercantile* [1994] FSR 275: crucial question is whether the idea is a general idea or a detailed idea — the former is not protected but the latter is

- *Cantor Fitzgerald Int'l v Tradition (UK)* [2000] RPC 95: if a substantial part of the programmer's skill and labour resides in the algorithm, may be protected

NB. Copyright protection of an algorithm only goes to the way the algorithm is expressed ie. the procedural steps set out in natural language.

NB. Implementing an algorithm in the form of a computer program is use of the algorithm (not copying)!

Q: What should one do to prevent his algorithm from being implemented?

Criteria for Protection

- as a literary work, a program must be
  - original (see lecture 1); AND
  - fixed (satisfied by programs stored on hard disk, CD, USB, magnetic tape etc)

NB. Programs in main memory are also sufficiently fixed (analogous to writing down on paper but erased later).

E1: X implements binary search on an array based on a flow chart in a book. X's program copyrightable?

E2: X has written several objects for drawing straight lines, ellipses, rectangles and triangles. Later Y incorporates these objects into his CAD program. Y's program copyrightable?

Authorship

- author is the person who creates the work: s.11(1)

- collaborative works — two cases:
  1. *distinct* authors
     - » contribution of each separable from that of others (eg. each writes one module of a program) — each is the author of his own part

2. *joint* authors
   - » contribution of each not separable from that of the others — all are "joint authors" of the entire work: s.12(1)

NB. Contributions of an author must not be trivial and must go to the expression of the work.

NB. Skill and labour in testing a program not sufficient for authorship: *Fylde Microsystems v Key Radio Systems* [1998] FSR 449.

Computer-Generated Work

- work generated by computer ***in circumstances such that there is no human author*** of the work: s.198(1)

Q: Are the following computer-generated?

- design drawings produced by CAD program
- diagnosis produced by medical expert system based on patient's input
- output of ChatGPT

- s.11(3): author of computer-generated literary, dramatic, musical or artistic work = "person by whom the arrangements necessary for the creation of the work are made"

Q: Who is this person?

Ownership of Copyright

- author is 1st owner: s.13
- collaborative works — two cases:
  1. *distinct* authors
     - » each owns the copyright in his own part
  2. *joint* authors
     - » all authors own the copyright in the whole work jointly in equal shares as "tenants in common": *Lauri v Renad* [1892] 3 Ch 402; *Stuart v Barrett* [1994] EMLR 448.
     - » but any act requiring licence requires the licence of all: s.194(2)

- owner of computer-generated literary, dramatic, musical or artistic work also governed by s.13

NB. OpenAI (developer of ChatGPT) Terms of Use:

> **[Ownership of Content] "As between you and OpenAI, … you (a) retain your ownership rights in Input and (b) own the Output."**

See, openai.com/policies/terms-of-use.

NB. DeepSeek Terms of Use:

(1) "You retain any rights, title, and interests - if any - in the Inputs you submit."

(2) "We assign any rights, title, and interests - if any - in the Outputs of the Services to you."

See, https://chat.deepseek.com/downloads/DeepSeek%20User%20Agreement.html.

Employee work

- work made in the course of the author's employment — copyright belongs to the author's employer subject to any contract to the contrary: CO, s.14(1)

- 1st question: "employee" or "independent contractor " ie. free-lance?
  - not difficult to answer in computing field

- 2nd question: "in the course of employment" is a question of fact
  - translation by a journalist done in his own time and employer had paid him a separate fee: *Byrne v Statist* [1914] 1 KB 622
  - lectures given by an accountant for which employer had provided secretarial help and paid expenses in relation to the lectures: *Stephenson Jordan v MacDonald* (1952) 69 RPC 10

Commissioned work

- work developed by self-employed author on commission — ownership governed by contract between the author and the commissioner: CO, s.15(1)

- court may infer an implied agreement on ownership in circumstances suported by clear evidence: *John Richardson Computers v Flanders* [1993] FSR 497

- even if no agreement (express or implied) on ownership, commissioner has 2 statutory rights:
  - an exclusive licence to exploit the commissioned work for all purposes that could reasonably have been contemplated by him and the author at time of contract; and
  - power to restrain any exploitation of the work for any purpose which he could reasonably object: CO, s.15(2).

E1: X, computing officer of T Ltd, writes a program to manage the company's inventory. Owner of the program?

E2: As in E1. X writes a program during office hours using company resources for his girlfriend.

E3: Y, office assistant of the CS Dept, writes a program monitoring the activities of the CS lecturers.

E4: Z, a free-lance programmer, contracts with V Ltd and writes an auditing program for the latter.

Term of Protection

- for literary, dramatic, musical and artistic works, term = life of author plus 50 years (expires on 31 December of the relevant year): CO, s.17(2)

- for works of "joint authorship", reference to "author" is to the last surviving author: CO, s.17(7)

- term of copyright in computer-generated literary, dramatic, musical or artistic works = 50 years from making: s.17(6)

E1: X and Y jointly wrote a piece of software which was published in 2000. They later died on 14 April 2003 and 28 August 2005 respectively.

E2: Windows 2000 was completed in 2000. Expiration date of copyright?

Rights of Copyright Owner ("Restricted Acts"): s.22(1)

1. copy the work: s.23
2. issue copies to the public: s.24
3. rent work to the public: s.25
4. perform, show or display work in public: s.27
5. communicate to public: s.28A
6. make adaptation: s.29

7. do any of the above in relation to adaptation: s.29(2)

NB. Doing, or authorising another to do, any of the restricted acts without licence infringes copyright: s.22(2).

NB. All these are *primary* infringements, which do not require proof of guilt in the infringer's mind and generally incur only civil liabilities.

NB. Suffice for infringement if the restricted act is performed in relation to a substantial part of the plaintiff's work: s.22(3).

NB. Infringement can be direct or indirect and it is immaterial whether any intervening acts themselves infringe copyright: s.22(3) ("causal connection").